Shift or replenishment? Reassessing the prospect of stable Spanish bilingualism across contexts of ethnic change

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Abstract

Much of the existing literature on Latinos' use of Spanish claims that a general pattern of intergenerational decline in the use of Spanish will produce an overall shift away from Spanish use in the U.S. (Rumbaut, Massey, and Bean 2006; Veltman 1983b, 1990). In contrast, recent works emphasize the importance of the social and linguistic context in reinforcing the use of Spanish as well as (pan)ethnic identities among U.S.-born Latinos (Linton 2004; Linton and Jiménez 2009; Stevens 1992). This literature suggests conditions under which Spanish-English bilingualism might become stable at the level of metropolitan areas; however, such conditions depend on how immigration shapes the context of language use for native-born Latinos. Given the declining levels of immigration from Latin America, will bilingualism subside in the U.S., or have certain communities created conditions in which bilingualism can be stable? Using geocoded data from restricted access versions of the Survey of Income and Program Participation (SIPP) and the American Community Survey (ACS), we model the probability of Spanish-English bilingualism among second- and third-generation Latinos using multilevel models with contextual measures of immigration and language use at both the neighborhood and metropolitan levels. We find evidence that U.S.-born Latinos are heavily influenced by the prevalence of Spanish use among U.S. born Latinos at both the metropolitan and neighborhood levels. Further, the proportion of foreign-born Latinos has little effect on the native born, after controlling for Spanish use among U.S,-born Latinos. These results are a first step in understanding the link between ethnic or panethnic contexts and language practices, and also in producing a better characterization of stable bilingualism that can be tested quantitatively.

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Introduction

Will the future of Latino ethnicity in the U.S. be the maintenance of country-of-origin ethnicities, the consolidation of a common panethnic identity, or a gradual process of incorporation marked by loss of ethnic or panethnic identities? As noted by a number of authors over the past five decades, an important part of the answer to this question revolves around the degree of Spanish language maintenance and use, as it is a significant component and marker of ethnic identity (Fishman, Ma, and Cooper 1975; Jiménez 2010; Linton and Jiménez 2009; Portes and Rumbaut 2001; Portes and Schauffler 1994; Telles and Ortiz 2008; Telles and Sue 2019). Although virtually all U.S. born Latinos speak English fluently, they often make remarks about the importance of the Spanish language to their (pan)ethnic identities like "Latino" or "Mexican American" (Garcia 2020; Telles and Sue 2019). Accordingly, Spanish proficiency can strengthen Latinx individuals' claims to (pan)ethnic authenticity (Garcia 2020; Jiménez 2010; Telles and Sue 2019). More importantly, ethnic and panethnic identities also influence Latinos' practices related to the preservation of Spanish (Bailey 2007; Garcia 2020; Hurtado and Vega 2004; Telles and Sue 2019).

Not surprisingly, 20.5 out of 35.8 million U.S. born Latinos report speaking Spanish at home, which makes for one of the largest Spanish-speaking populations in the world[1]. Not only has the number of Spanish speakers grown in recent decades, but also new forms of Spanish have emerged in the United States (Lipski 2008; Ramírez 1988; Zentella 2007).

[1] Data from the 2019 American Communities Survey.

Nonetheless, sociologists have long debated the trajectory of Spanish bilingualism. The literature on language shift has predicted the eventual disappearance of Spanish (Veltman 1983b, 1990). Empirical work seems to confirm that the Spanish language may succumb, over generations, to the American linguistic graveyard (Rumbaut et al. 2006). In contrast, recent literature has examined the role of social context in reinforcing and even stabilizing the use of Spanish language among U.S.-born Latinos, reopening the debate about the future of Spanish (Linton 2004; Linton and Jiménez 2009). In particular, the ethnic replenishment hypothesis posits that the continued flow of Latin American immigrants sharpens ethnic and panethnic boundaries that reinforce the use of Spanish among U.S.-born Latinos (Jiménez 2010; Linton and Jiménez 2009).

Whether their prediction is shift or retention, a shortcoming of the sociological analyses of bilingualism is their heavy reliance on demographic factors tied to the immigrant generation, such the steadiness or magnitude of immigrant flows, or the total number of Spanish speakers in a community. As these factors vary widely across American communities and the net flow of Latin American immigration is now zero, the overall path of bilingualism remains unclear. How can we identify the broad trajectory of bilingualism if there is heterogeneity in social processes and linguistic practices? Individual- and family- level characteristics may affect practices like the transmission of Spanish from the second to the third generation. Outside of the family, the density of fellow speakers varies widely across neighborhoods and cities. Given these factors, how is the intergenerational trajectory of Spanish affected by the neighborhood context, and what are the places in the U.S. where bilingualism might be stable? Further, can bilingualism be the result of decisions to preserve the Latino panethnicity or the ethnicities within?

We approach these questions with multilevel models of individual, family, and contextual determinants of bilingualism, including contextual measures and effects at both the metropolitan area and neighborhood (census tract) levels. Combining restricted-use geocoded data from the Survey of Income and Program Participation (SIPP) and the American Communities Survey (ACS), we test a hypothesis of stable bilingualism in a study design that can also account for the effects of generation-since-migration and the ethnic replenishment hypothesis. Our findings suggest that the ethnic replenishment hypothesis does not hold for second-generation and third-or-later generation Latinos. Instead, our findings suggest a process of peer influences at the metro area and neighborhood levels. Second-generation and third-or-later-generation Latinos are more likely to speak Spanish at home (our measure of bilingualism) as the number of native-born Spanish speakers in their neighborhood and metro area increases. This empirical pattern, in turn, supports a model of bilingualism that doesn't depend on the continual flow of large numbers of foreign-born Latinos, and suggests the conditions under which this bilingualism for native born Latinos might be sustainable over time.

Literature Review

Processes of language change: language shift and contextual reinforcements

Will the use of Spanish in the United States follow the overall pattern of decline observed among immigrant groups, or will the increasing Latino population maintain Spanish over time? The extent of Spanish use in the United States not only has been subject of debate among sociolinguists and demographers, but also it also has raised public concerns about the integration

of Latinos into American life[2] (Chavez 2013; Skrabanek 1970; Veltman 1981, 1988b). The trajectory of Spanish has been key in assessing whether the United States is an Anglo-dominant society (Skrabanek 1970) or a multilingual society where bilingualism can become stable (Fishman 1969b; Fishman et al. 1975). As evidence in earlier studies consistently pointed to the intergenerational decline Spanish bilingualism (Ramírez 1988), the "language maintenance or shift" framework ruled out the prospect of stable bilingualism (Grenier 1984; Veltman 1983b). Despite the overarching prediction of Spanish language loss, research on language maintenance or shift has identified factors that foster bilingualism, such as proximity to the U.S. border, family formation, or gender (Bills, Chávez, and Hudson 1995; Lutz 2006; Stevens 1985, 1992). Recent approaches have hypothesized contextual mechanisms that can reinforce the use of Spanish among U.S. born Latinos (Jiménez 2010; Linton 2004; Linton and Jiménez 2009b). For instance, the critical mass model by Linton (2004) posits that, as the number of Spanish speakers in a community increases, U.S.-born Latinos have a stronger incentive to adopt Spanish. Similarly, the ethnic replenishment hypothesis predicts that the continued flow of migration from Mexico and Latin America will sharpen ethnic and panethnic boundaries, producing incentives for bilingualism among native-born Latinos (Jiménez 2010; Linton and Jiménez 2009).

The language maintenance and shift framework established the importance of analyzing language change in social sites where different linguistic groups engage (Fishman 1964). This research originally focused its analysis on the shift away from foreign-language use, as an alternative to the "stable bilingualism" hypothesis (Fishman 1969b; Fishman et al. 1975;

[2] The "Veltman report," (1988b) commissioned by the Hispanic Policy Development Project, clearly illustrates public concerns about the Spanish language, as it provided reassurance that U.S.-born Latinos are fluent in English and on the path to English monolingualism.

Veltman 1981). Originally concerned with the adoption and proficiency of English among the children of immigrants, this literature ultimately focused on bilingualism due to the high rates of English-speaking among the children of immigrants (Veltman 1981, 1983a). As the empirical evidence pointed towards a decline in foreign-language use across immigrant generations (Ramírez 1988), the question of bilingualism became commonly framed as a process of "language shift" or "mother-tongue shift" (Bills et al. 1995; Stevens 1985; Veltman 1983b). Nonetheless, other studies along this line of research keep their focus on the maintenance of foreign languages (Fishman 1978; Lopez 1978; Mejías, Anderson, and Carlson 2002; Resnick 1988; Skrabanek 1970).

The language shift framework has made extensive use of a household model of language shift, in which there are successive drop-offs in Spanish use between immigrants and their children, and between the second and third generations. Overall, the language shift and maintenance framework posits that, as immigrant groups adopt the home country language, their use of foreign languages will become increasingly limited to intimate contexts like households or churches (Fishman 1964). To assess this, earlier research measured multiple dimensions of language use, such as proficiency in speaking or writing, across different domains of language use (e.g., household, workplaces, shopping, with friends) (Fishman 1969a; Ramírez 1988; Skrabanek 1970; Veltman 1983a).[3] Although foreign languages are used in a variety of situations, these earlier studies identified the household as the setting in which Spanish is most often used (Ramírez 1988; Skrabanek 1970; Veltman 1983b). For example, Skrabanek (1970) found that Mexican Americans of different age groups use Spanish more frequently when they talk to adults within their household. Veltman (1983a) ultimately characterized communication

[3] Some studies even have measures of preferred languages.

between immigrant parents and their children as the only relevant domain of foreign-language use, despite having data that showed some degree of foreign-language use outside of households. Veltman also established that the third generation tends towards English monolingualism, since their U.S. born parents are already proficient in English (Veltman 1983a, 1988a). Thus, Veltman's depiction of the language shift effectively captures the "generation clock" process of a straight-line assimilation model. The parameters of Veltman's (1990) language shift model formed the basis for his prediction that Spanish would eventually subside when immigration from Latin America stops.

Evidence from quantitative studies tends to confirm the decline in bilingualism across immigrant generations. Survey-based studies of different country-of-origin groups in communities in Texas, Colorado, New York City, and New Mexico consistently show declining rates of Spanish use for 2nd and 3rd generation Latinos (Ramírez 1988). Arriagada (2005) finds declining rates of Spanish bilingualism for second and third generation Latino children, net of detailed household and individual characteristics. Rumbaut, Massey, and Bean calculated the "life expectancies" of foreign languages and found that bilingualism rates decline for all language groups across immigrant generations, and that the life expectancy of foreign languages is 2.5 generations, on average (2006). Despite the prevalence of the intergenerational drop-off, Spanish language has the higher retention rates among the second and third generation, when compared to other linguistic groups (Rumbaut et al. 2006b; Thomas 2011).

Prior evidence appears to generalize the trend towards language loss and validate the intergenerational household loss as the key mechanism behind the shift. Nonetheless, the language shift and maintenance approach has analyzed factors that favor bilingualism, including the role of social context. Stevens (1985) found that the children of two foreign-born parents are

the most likely to learn the foreign language of one of their parents, but this happens only among groups with high rates of linguistic ethnic homogamy. Household composition is another important source of language retention. The use of Spanish by family members (parents, grandparents, and siblings), as well as the presence of grandparents and other adult relatives within households contributes to the retention of Spanish among Latino children (Arriagada 2005). Family formation and composition can also depend on context, as residential segregation of ethnic groups produces higher levels of linguistic endogamy, which reinforces the use of foreign languages (Stevens 1992). These studies, directly or indirectly, still emphasize the household as the key linguistic domain for language retention. Other studies, however, find significant effects of contextual factors, such as proximity to the U.S. southern border or the concentration of coethnics in the neighborhood (Bills et al. 1995; Lutz 2006).

Recent approaches have deviated from the language shift perspective to emphasize contextual mechanisms, beyond the household and family environments, that can reinforce bilingualism and even stabilize it. Building on the network mechanisms of "critical mass" and "tipping" models, Linton (2004) posits that, as the number of Spanish speakers in a community increases, Latinos have a stronger incentive for and lower costs from adopting Spanish. Linton finds that U.S.-born Latinos in metropolitan areas where the percent of Latino bilinguals is greater than a "critical mass" threshold are 50% more likely to be speak Spanish at home[4] (2004). The probability of speaking Spanish also increases with the percent of foreign-born Latinos in the metropolitan area (Linton 2004). As there is sizable geographic variation in the rates of Spanish use, these findings highlight the role of context in promoting bilingualism at the

[4] The critical mass is defined as "the average level of bilingualism in places where more than half of the native-born/1.5 generation population is bilingual. It is assumed that contextual incentives in these places have "tipped" in favor of bilingualism" (Linton 2004:298).

individual level (Linton 2004). Nonetheless, metro area-level context measures identify a broad and even distal social environment. This type of analysis can be improved by measuring contextual factors at the neighborhood level. Additionally, the contextual measures used by Linton (2004) put the weight of language maintenance on the relative size of the immigrant community. As the U.S. born Latino population has grown and spread geographically, network and peer effects hypotheses can be refined to distinguish between the contextual effects of foreign-born-and U.S- born- Latinos on individual native-born Latinos' language practices.

The replenishment hypothesis posits that the continued flow of migration from Mexico and Latin America is sharpening ethnic and panethnic boundaries (Jiménez 2010; Linton and Jiménez 2009). Immigration can sustain or increase the number of Spanish speakers in a community, resulting in Spanish language preservation among native-born Latinos as part of the ethnic replenishment process (Linton and Jiménez 2009). Although this framework was first developed around Mexican American ethnic boundaries (Jiménez 2010), Linton and Jiménez (2009) extended the argument to analyze bilingualism as the result of a replenished Latino panethnicity. They test the replenishment hypothesis with an OLS regression of the rate of bilingualism among U.S.-born and 1.5 generation Latinos in select metropolitan areas as a function of the proportion of foreign-born Latinos in the metro area and other contextual measures. The theorized mechanism behind this operationalization is that bilingualism can become stable as immigrants "replenish" the "identity repertoire" of U.S. born Latinos, through intervening mechanisms of social contact between immigrants and 1.5-or-later-generation Latinos (Linton and Jiménez 2009:985).

The replenishment hypothesis opens possibilities for the analysis of contextual effects indicative of social contact beyond endogamy (which ultimately defines household interaction).

Nonetheless, we offer a number of reasons why the conceptualization and testing of the replenishment hypothesis, as a driver of stable bilingualism, should be revisited and refined. First, the conceptualization of replenishment as social contact between immigrant and U.S. born (and 1.5 generation) Latinos is largely contingent on immigration. As the overall migration flows from Latin America are on decline, this process could eventually fail to produce stable bilingualism. A stronger sign of stable bilingualism through social contact and peer effects could come from measures that reflect that U.S.-born Latinos influence one another. Second, although the geographic variation in demographic patterns can help identify metro areas where bilingualism is stable, measuring replenishment as the change in the proportion of foreign-born Latinos (Linton and Jiménez 2009) has pitfalls. Such measurement may not fully capture the concept of replenishment (e.g., a continued influx of new immigrants over time), and it instead it could be picking up patterns of recent immigrant arrival (e.g. "new destinations") and even out-migration of foreign-born Latinos. Finally, the test of the replenishment hypothesis was done only at the aggregate level (Linton and Jiménez 2009), which complicates the interpretation of these results as evidence of replenishment through social contact outside of the household. As previous research has consistently signaled the relatively high contribution of the household roster towards the use of Spanish at home, a proper test of the replenishment hypothesis should account for immigrants or Spanish speakers in the household roster in conjunction with any plausible contextual measure of ethnic replenishment. This step is key in testing ethnic replenishment, because the aggregate test may be ultimately reflecting the presence of immigrant multigenerational households in metropolitan areas.

Latinos' (Pan)ethnic identities and language practices

As we noted previously, the future of Spanish bilingualism in the U.S. is conceptually tied to Latino ethnic and panethnic identity formation and change, as the Spanish language is a salient marker of ethnic identity and authenticity (Fishman et al. 1975; Garcia 2020; Jiménez 2010; Portes and Rumbaut 2001; Telles and Sue 2019). Although the emergence of a Latino panethnicity could be partially attributed to the size and continuity of Latin American immigration between 1965 and 2010, such demographic pressures may not be enough to understand the connection between the Spanish language and the ethnic and panethnic identities of Latinos.

The Spanish language is salient and valuable to those who claim a Latino identity, beyond the practical benefits of retaining its use (Garcia 2020). Regardless of their individual Spanish proficiency, U.S. born Latinos typically acknowledge the importance of knowing Spanish towards claiming identities such as "Latino" or "Mexican American" (Garcia 2020; Telles and Sue 2019). Typically, the Spanish language has been characterized as a marker of (pan)ethnicity; thus, if the language is maintained, an ethnicity is maintained. However, this relationship could also act in the opposite direction. The "Latino" panethnic self-identification by itself can act as distinct "situational pressure" that motivates second-generation Latinos to cultivate bilingualism among their third-generation children, independently from other contextual pressures (Garcia 2020). In this study, we will test whether measures of ethnic preservation can have an effect on Spanish language use.

The connection between ethnic identity and the use of Spanish is complex and heterogeneous. The Spanish language is not always coterminous with or essential to Latino (pan)ethnic identities. Mexican Americans claim and value their identities as "Latino" or

"Mexican American" even when they don't speak Spanish (Telles and Sue 2019). The use of Spanish does not represent a linear path to ethnic or panethnic solidarity among Latinos. Instead, knowledge of Spanish in general and of specific Spanish dialects in particular opens up situational options for "ethnically flexible" Latinos to signal their panethnic or ethnic belonging through conversational cues that are part of their "ethnic repertoire" (Negrón 2011). A U.S.-born Latino of a given ethnicity can deploy such cues to seek collaboration with Latinos of a different ethnicity (Negrón 2011). Latinos' bilingualism increases options in coethnic and inter-ethnic settings, enabling strategies such "accommodation without assimilation" (Linton and Jiménez 2009b; Mouw and Xie 1999).

Racialization also complicates the way in which Latinos feel about Spanish and use the language, because racialization creates pressures for and against bilingualism. The Spanish language has been a site of contestation about the place of Latinos in American society (Aparicio 1998; Huntington 2004). On one hand, American society has presented Latinos with social and institutional pressures to speak only English (Aparicio 1998). On the other hand, non-Latinos (whites in particular) impose their expectations on Latinos to know Spanish (Telles and Sue 2019). The optional and contested qualities of bilingualism mean that not all individuals will have the same connection to the Spanish language, compromising the stability of bilingualism.

Ethnographic work has identified a number of factors that account for variation in the language practices of Latinos, including "generation clock" effects. Although scholars of sociolinguistics and Latino studies have been critical of the inter-generational processes outlined in the frameworks of assimilation and language shift, many works acknowledge the increase in English monolingualism and decline in bilingualism across generations (Bailey 2007; Hurtado and Vega 2004; Negrón 2011; Potowski 2015). Nonetheless, the language shift may not always

immediately impact the second generation and instead it could be more noticeable by the third or fourth generation (Bailey 2007).

In terms of ethnic options, a language shift in the second and third generation is not always deleterious to the Latino panethnic identity (Negrón 2011). Increased use of English between Latinos may loosen the association between distinct Spanish variations and ethnic identities (Potowski 2015). Further, a degree of language shift enables Latinos to cross over from ethnic to panethnic identities (Negrón 2011). More importantly, change in linguistic practices across generations doesn't always involve a loss of the Spanish language, as Garcia finds that bilingualism may change in form for second-generation Latinos, but not in intensity (2020). Thus, while it is still important to account for differences in Spanish use across immigrant generations and anticipate a pattern of decline, we still need to analyze whether and how such patterns vary.

Along with the process of incorporation and ethnic identity, the language practices of Latinos are also influenced by class, gender, and the life course. This occurs in unison with Spanish dialect differences between ethnic groups, which open up ethnic and linguistic options (Negrón 2011; Potowski 2015; Zentella 2007). The class connotations of some Spanish dialects that are considered inferior vis-a-vis cultured or standard forms of Spanish can be a source of "linguistic insecurity" among the children of immigrants (Zentella 2007). This is related to prescriptive, middle-class practices that enforce the use of "proper" forms of Spanish and warn against the use of dialects and word choices that are considered lower class (Telles and Sue 2019; Zentella 2007). The use of Spanish can fluctuate throughout the life course. Although many U.S.-born Latinos don't become Spanish-proficient during childhood, their exposure to Spanish in early life creates "linguistic bands" that enable them to learn the language later in life

(Hurtado and Vega 2004). For example, second-generation Dominicans in Rhode Island re-learned Spanish or increased their use of the language as they undertook new roles in work and family settings, which consequentially "enmeshed many in Spanish-speaking social networks and positions in which there is little ambiguity of identity" (Bailey 2007:160).

The character of cities, neighborhoods, or regions can also reinforce bilingualism. Communities like San Antonio in Texas show signs of being a "Spanish-language context" with a long standing Spanish-language culture, which is often acknowledged by Mexican American residents, and which may account for higher rates of Spanish proficiency than what can be explained by relevant measurable characteristics (Telles and Ortiz 2008:196; Telles and Sue 2019). Proximity to the U.S. border or to large Latino urban centers is also associated with higher Spanish retention among Latinos (Bills et al. 1995). Since community-level processes may provide enough linguistic pressures or critical-mass conditions to foster bilingualism, it is important to explore the geographic heterogeneity in language practices and language preservation.

Hypotheses

Table 1 provides an overview of the competing perspectives on bilingualism and language shift discussed above and included in the empirical tests presented in our analysis. Language shift appears as a consistent feature of quantitative and qualitative studies, and it's operationalized by the effect of generation-since-migration on bilingualism outcomes. Ethnic replenishment is a plausible mechanism that requires full consideration, as many communities in the U.S. have become destinations for Latin America immigrants. The replenishment hypothesis is tested with the effect of measures of migration flows on the outcomes of native-born Latinos. As discussed

above, ethnic replenishment depends heavily on immigration, which makes it a precarious source of stable bilingualism. The third perspective under consideration is labeled "stable bilingualism." Our operationalization of stable bilingualism aims to capture processes that reinforce bilingualism beyond immigration. Here we offer a revised, testable hypothesis to identify stable bilingualism and compare it to the empirical test of ethnic replenishment and the operationalization of the language shift as a result of the intergenerational decline in bilingualism (Table 1). These different hypotheses don't necessarily exclude each other, as the decline in bilingualism across generations can occur despite the presence of contextual reinforcements.

Our operationalization of the "stable bilingualism" hypothesis has two possible types of mechanisms (Table 1). We posit that a) U.S.-born Latinos may influence each other, whether due to social contact among bilingual U.S. born Latinos at the local level, or through peer or network effects. Such network effects may take the form of the "critical mass" model by Linton (2004), although we maintain that it is important to analyze the influence of immigrant and native Spanish speakers separately in order to identify a form of stable bilingualism that doesn't depend on the presence of immigrants. We also argue that b) there could be "community effects" characterized by a community identity that values Spanish, driven by ethnic or cosmopolitan cultural values. A community effect could be measured indirectly through geographic-level random effects, after controlling for a set of contextual measures related to Spanish language use. We also present a strong version of our stable bilingualism hypothesis, in which the effects of the "generation clock" become small or negligible. This would involve a strong intergenerational transmission of the Spanish language among Latino families. The strong version of stable bilingualism, in general, could be the result of effects of contextual characteristics and even a "community effect" in the language practices of families. Hence, the strong version of stable

bilingualism could reflect selective acculturation to some degree, as some ethnic communities have favorable modes of incorporation and the ethnic resources that enable the cultivation of Spanish use by local ethnic institutions and families in an ethnic community (Portes and Rumbaut 2001). The following section outlines how operationalize and test both the ethnic replenishment hypothesis and the peer effects mechanism of our stable bilingualism hypothesis.

[Table 1 about here]

Data and Methods

In this paper, we use a combined data set of restricted access data from the American Community Survey (ACS) and the 2014 panel of the Survey of Income and Program Participation (SIPP) to estimate multi-level models of the immigrant generation process and community-level contextual effects on Spanish language bilingualism for a sample of native-born Latinos. The versions of the ACS and the SIPP that we use are available at the Federal Statistical Research Data Centers (FSRDC) and include geocodes for respondent's Census tracts, which enables us to create measures of language use at detailed levels of geography in order to estimate local neighborhood and broader spatial effects on Spanish language maintenance.

Table 2 provides a description of the variables that are used in the analysis. In our analysis, we use data from the 2014-2018 ACS to create contextual variables measuring the prevalence of Spanish language use at the neighborhood and metropolitan level in order to estimate models of the broader pattern of spatial dynamics. In addition to measures of household-level relationships, including multigenerational links between the foreign and native born, we hypothesize that Census tract level neighborhood characteristics will be a key indicator

of Spanish language use among native born Latinos. A key benefit of use of our geocoded SIPP and ACS data is that it allows differentiate between neighborhood and metro-level contextual effects on bilingualism. Rows 2-3 of Table 2 describe variables measuring the proportion of the population who are native- and foreign-born Latinos, respectively, at the metro level. Note that what we are calling a "metro" level is actually covers both metropolitan areas and rural counties: For urban areas, the ACS data is aggregated at the metropolitan area, and in rural areas the data is aggregated at the Public Use Micro Area (PUMA) level, which represents a county or a combination of contiguous counties in sparsely populated areas. Rows 4 and 5 provide corresponding measures of the relative size of the Latino population at the Census tract level. In contrast, Rows 6 and 7 of Table 2 describe two variables measuring linguistic contextual effects based on the proportion of native-born Latinos who speak Spanish at home at the tract and metro level. These variables don't measure the size of the Latino population, but focus on language behavior in the broader social environment as a possible "peer effects" encouraging Spanish language.

[Table 2 about here]

In addition to the use of the ACS to create contextual level geographic variables, we use the SIPP as the source of individual-level cases for our analysis. An important feature of the 2014 SIPP is that it includes a question on parental nativity, which allows us to distinguish first- and second- generation immigrants from native-born respondents with native-born parents. Aside from the Annual Demographic Supplement in the March Current Population Survey (CPS), the SIPP is the only large demographic data set that includes a measure of parental

nativity. Using the geocodes available in the restricted access version of the SIPP data, we link respondents to the ACS level geographic data.

In our statistical models, we use hierarchical multilevel models to estimate the effect of neighborhood, metro, and regional effects on Spanish language use among second and third-plus generation Latino respondents in the 2014 SIPP. An important feature of these models is allowing the effects of individual- and household-level characteristics to be estimated as random coefficients that vary across areas. For example, if the effect of geographic context—such as the density of Spanish language speakers at the labor market and regional level--alters the parameters under which language use and maintenance is taking place, then this may be evident in variation in the effect of individual-level factors such as generation-since-immigration on the level of Spanish language use.

[Maps 1-3 about here]

[Tables 3-4 about here]

Results

Maps 1-3 provide a descriptive overview of the geographic variation in basic demographic measures related to language use. First, Map 1 shows the size of the Latino population, as a basic indicator of spatial concentration in certain areas that are likely to be hospitable to the long-term maintenance of Spanish-English bilingualism among the native-born population. In contrast, Map 2 shows a different measure, the proportion of native-born Latinos who speak Spanish at home by Public Use Micro Area (PUMA), which closely corresponds to our key independent variable of metro-level native born Spanish prevalence discussed in Row 5 of Table 2. It is notable that this map is not perfectly correlated with the spatial concentration of Latinos

per se in Map 1. While the high proportion of native-born Spanish speaks in southern Texas and Florida makes sense, it is also clear that the proportion is high in new destination areas as well, such as the Carolinas and scattered counties through the Midwest, which, on the surface, might seem puzzling. Some of can be explained by Map 3, which shows the proportion of the Latino population that is foreign born, indicating higher immigrant proportions in many new destination areas outside of established Latino regions of the country. One possibility, then, is that spatial variation in native born Spanish speaking from Map 2 can be explained by the areas with more "recently" arrived Latino populations—as indicated by the proportion foreign born—as native born Latinos in the ACS will be more likely to be second generation (as opposed to third generation +), and, perhaps most importantly, to be living with first-generation immigrants at home. The advantage of our SIPP analysis, as discussed above, is that it allows us to differentiate between 2nd and 3rd generation immigrants, allowing for a clearer test of the relative effect of social context and peer related linguistic factors. Nonetheless, Map 2 shows that there is a lot of spatial variation in the proportion of native born Latinos who speak Spanish at home both within and outside of areas with large Latino populations in Map 1; thus the question is what is the importance of these factors on language use among second and third generation Latinos in the SIPP net of control variables for education and household composition.

Table 3 shows results for multilevel logit models of the probability of speaking Spanish at home for second generation Latinos in the SIPP, and Table 4 presents results for third generation Latinos. Because of the native-speaker fluency in English is almost universal among native born Latinos, we argue that speaking Spanish at home is equivalent to evidence of bilingual language ability among native born Latinos. At the same time, this measure will underestimate Spanish language ability among Latinos as it possible to be a Spanish-English

bilingual but not speak Spanish at home; as such this reflects the limitations of the SIPP and ACS data in measuring the prevalence of Spanish language use and bilingualism, as discussed above. All models control for education and household composition based on the count of the number of people in the household (excluding the respondent) by generation, gender, and age. For example, in Model 1 of Table 3, we find that the number of first-generation adult (age≥18) women in the household increases the log odds of the respondent speaking Spanish by 0.561 (s.e. 0.197, p<.001), while, in contrast, the number of third generation male children (age<18) has no effect (b=0.0237, s.e. 0.0588).

For both Table 3 and 4, the results for the ACS social and linguistic contextual variables are presented in rows 1-6, and the main theoretical tests from Table 1 revolve around the relative importance of the size of the immigrant Latino population compared the size and language use of the native- born Latino population. Dividing the contextual variables into measures at the metro and neighborhood level also allows us to test whether language use is primarily a function of the clustering of Latinos within the micro-level environments of neighborhoods or a broader result of the social and economic context at the metro and PUMA level. If, for example, the language use of native born Latinos is largely the result of neighborhood contextual factors, then one might argue that stable bilingualism could exist in areas where Latinos are highly segregated at the neighborhood level even Spanish use is low overall at the metropolitan or regional level.

An key result in both Table 3 and 4 is that there is no evidence that the metro level proportion foreign born Latino (Metro %FB Latino) has a positive effect on the Spanish language use of native-born Latinos in any of the models. In addition, while the tract level proportion foreign born Latino has a positive effect on Spanish language use for third generation Latinos in Model 5, the effect becomes statistically insignificant at the p<.05 level in Model 6

after adding the native-born peer effect variables. We argue that this indicates a lack of support for the ethnic replenishment perspective, as there is no evidence that the neighborhood or metro level presence of immigrant Latinos is positively associated with higher rates of Spanish language use among either 2nd or 3rd generation Latinos, once we control for key measures of household structure and the size and language use of the native born Latino population.

Discussion and Conclusion

Our stratified analyses suggest that U.S.-born Latinos are more likely to speak Spanish when there are other U.S.-born Latinos who speak Spanish in their communities. The presence of native-born Spanish-speaking Latinos at both the metro area level and the neighborhood (census tract) level has positive and significant effects on the probability that a U.S. born Latino speaks Spanish at home. We found these effects for our second generation subsample, as well as for our third-or-later generation subsample. Since the presence of fellow U.S.-born Latinos matters for individual Spanish preservation, we conclude that context is indeed relevant for Spanish bilingualism. More importantly, native-born Latinos are influenced by fellow U.S.-born Latinos, regardless of the presence of foreign-born Latinos. This peer effect mechanism is not predicated on immigration as a source of language maintenance and can be a first sign of stable bilingualism at the local level.

Based on previous literature, we gave consideration to the ethnic replenishment hypothesis by including the prevalence of foreign-born Latinos in the metropolitan area and neighborhood. The ethnic replenishment hypothesis (Jiménez 2010; Linton and Jiménez 2009b) argues that the steadiness of immigration sharpens ethnic boundaries and influences the preservation of ethnic markers like the Spanish language among the native born. Our models,

however, show that native-born Latinos use of Spanish at home is influenced by the local presence of the native born rather than the foreign born. Moreover, it's the use of Spanish by other U.S.-born Latinos in the metro area and neighborhood that predicts a higher probability of individual Spanish use. Although the replenishment hypothesis is plausible and it may manifest in multiple markers of ethnicity, our results favor peer effects over replenishment as a contextual reinforcement of bilingualism.

Our peer effects finding is net of household-level composition by sex and immigrant generation. Our characterization of the household roster should capture well-known family dynamics that affect bilingualism, such as Spanish transmission in multi-generational households, parental transmission from first and second and from second to third-or-later generation Latinos, and couple formation. Thus, our findings are likely to not be an artifact of the prevalence of immigrant families or the rates of intermarriage at the local level. Controlling for household composition strengthens the interpretation of our findings as a peer effect.

The analyses presented here still leave two of our research questions open for further investigation. In our effort to identify stable bilingualism, at least at the community level, we posit that a strong version of this hypothesis involves the magnitude of the intergenerational drop-off in Spanish language use. To test this hypothesis, we will focus on the intergenerational change in Spanish use between the second and third-or-later generations. In places where the rates of Spanish use among the second generation and the third-or-later generations are comparable, the prospect of stable bilingualism may be more viable. Although we acknowledge the intergenerational drop-off in Spanish use as a feature of language change in the U.S., this process can vary widely across geographies, as some communities have formed an identity tied to the Spanish language (Telles and Ortiz 2008; Telles and Sue 2019). As a next step, instead of

stratifying our sample by generation, we will model the change from the second generation to the third-or-later generation on the full sample of native-born Latinos. To capture geographic heterogeneity and identify the places where the drop-off is minimal or zero, we will use random coefficient models of the "generation clock" effect, specifying such effect as a function of contextual variables and a community-specific error term. As a methodological improvement in our analysis, we will pool the 2014 with the 2018-2021 SIPP samples in order to increase the sizes of our subsamples of Latinos by immigrant generation, as well as to increase the power of our study at the neighborhood level.

Finally, we posed a question about the impact of ethnic change on bilingualism. For this purpose, we intend to request access to the information on names from the American Community Survey, in order to develop measures of naming patterns among Latinos. We argue that decisions about children's names are also part of the process of incorporation and ethnic preservation. The decision to maintain distinctly ethnic names may be an additional sign of the preservation of ethnic or panethnic identities. Our goal is to produce a valid measure of ethnic retention based on naming patterns at the metro area and neighborhood levels, to test whether these measures would predict higher probabilities of Spanish use at home for native-born Latinos in the SIPP.

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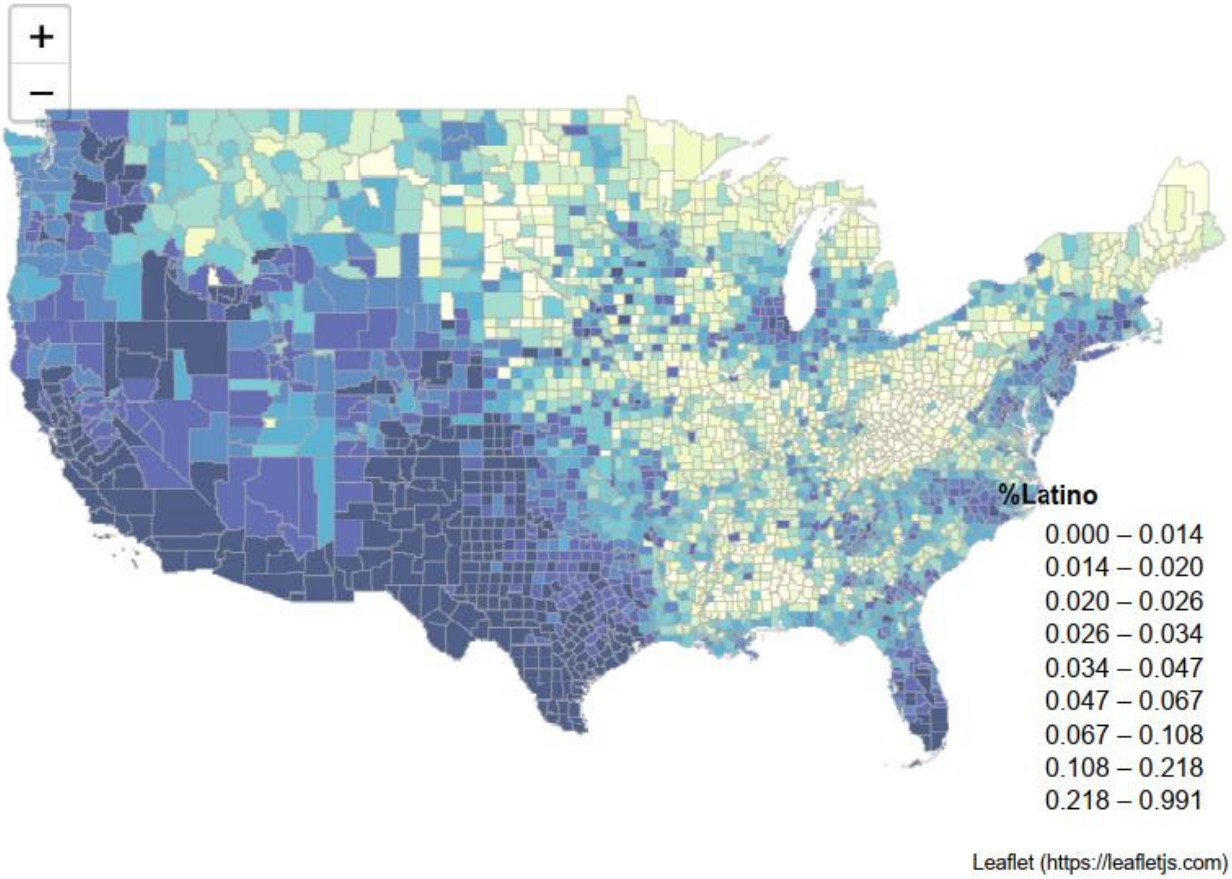
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Table 1: Competing Perspectives and Tests

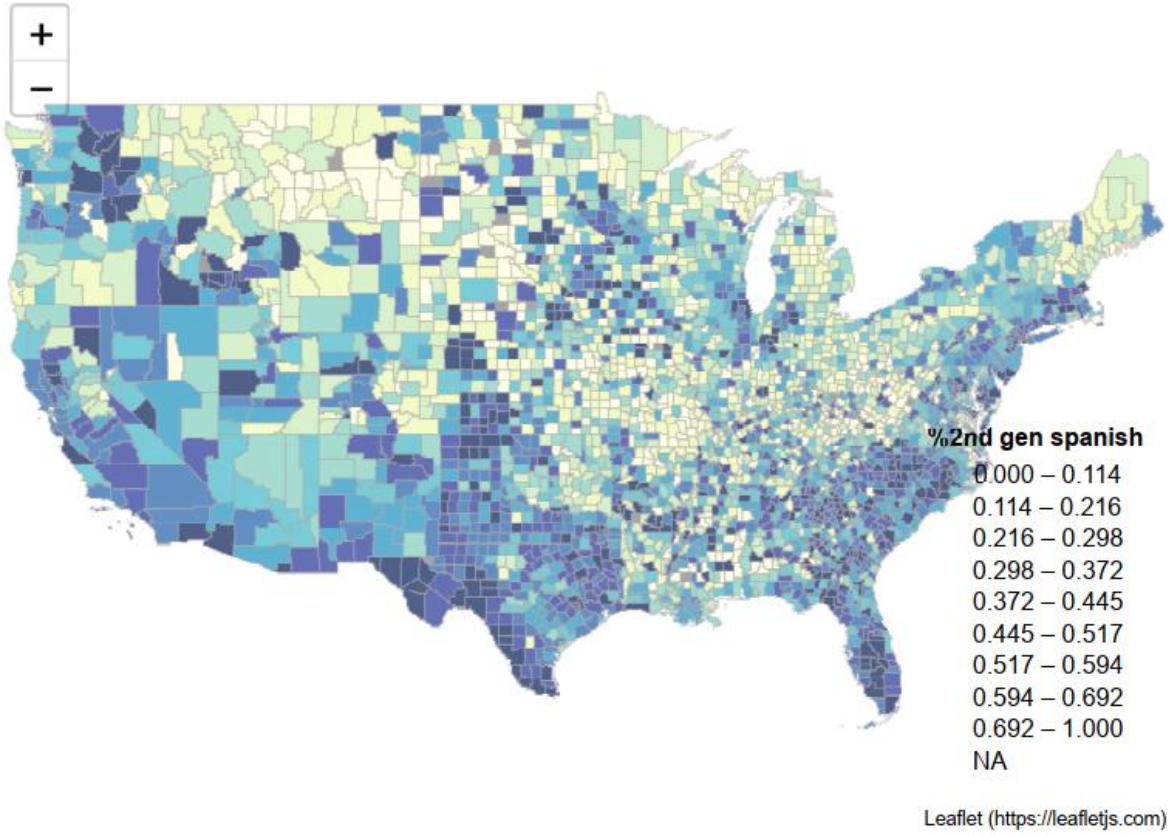
Perspective	Mechanisms	Test
Language Shift	Decline in transmission of Spanish from parents to children over immigrant generations	Generalized pattern of negative effects of generation-since-migration ("generation clock" effects) on individual bilingualism
Ethnic Replenishment	Continued flows of immigrants foster social contact between foreign-born Latinos and 1.5-or-later generation Latinos	Significant positive effects of a contextual measure of replenishment (i.e., continued flow of immigrants) on individual bilingualism
Stable Bilingualism	a. Social contact between U.S. born Latinos. Peer and/or network effects b. "Community effect," i.e., the community has a strong identity tied to Spanish (ethnic or cosmopolitan), often with institutional support like Spanish instruction in public schools	a. Positive effects of contextual rates of bilingualism among the U.S. born on individual bilingualism outcomes[1] b. strong, positive, community-level random intercept, net of individual, household, and contextual measures
	Strong version: Minimal or zero decline in rates of bilingualism between the second and third generation (i.e., reduced or negligible "language shift")	Minimal generation clock effects at the community level, identified with random coefficient models of the generation clock effect as a function of community effects and contextual measures

[1] Testing for peer effects may lead to further studies with datasets that collect network data.

Map 1: Proportion Latinos by PUMA, public use 2019 American Community Survey data.



Map 2: Proportion of native-born Latinos who speak Spanish at home by PUMA, public use 2019 American Community Survey data.



Map 3: Proportion of the Latino population that is foreign born by PUMA, public use 2019 ACS data.

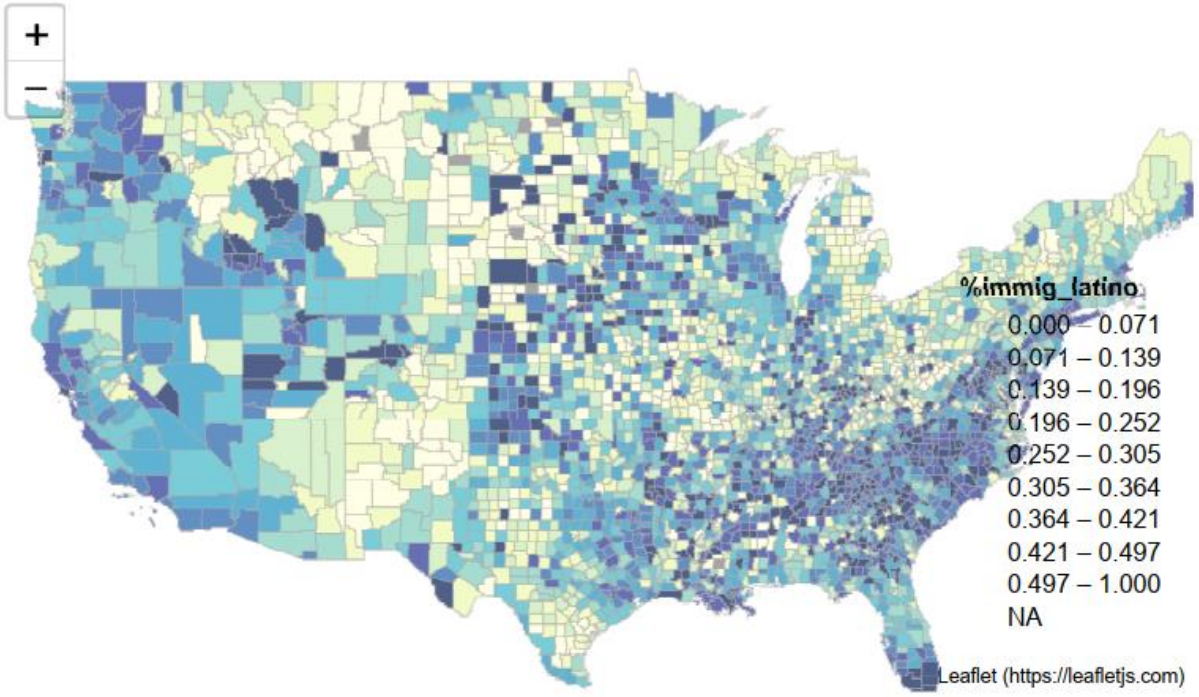


Table 2: Description of the variables in the model

Row	Variable name	Description
1	Bilingual (dependent variable)	Spanish-English bilingual is coded as 1 if the respondent speaks Spanish at home (source: SIPP).
	Linguistic context variables (from aggregate ACS data):	
2	Metro[1] level proportion native born Latino ("Metro %NB Latino")	The proportion of native-born Latinos in the metro area (or rural PUMA).
3	Metro[1] level proportion foreign born Latino ("Metro %FB Latino")	The proportion of foreign-born Latinos in the metro area (or rural PUMA).
4	Tract level proportion native born Latino ("Tract %NB Latino")	Corresponding measure for Metro %NB Latino at the tract level.
5	Tract level proportion foreign born Latino ("Tract %FB Latino")	Corresponding measure for Metro %FB Latino at the tract level.
6	Tract level proportion native born Latinos who speak Spanish at home ("NB Spanish Tract")	The proportion of native-born Latinos in the tract
7	Metro[1] level proportion native born Latinos who speak Spanish at home ("NB Spanish Metro")	Metro level equivalent of NB Spanish Tract
	Household level variables (from the SIPP):	
8	Household maximum education: high school	The highest level of education in the household is high school.
9	Household maximum education: college+	The highest level of education in the household is college or higher.
10	Household composition variables: counts by generation, gender, and age.	These variables are the number of people in the household (excluding the respondent) by generation, gender, and adult (18+) or child (less than 18).

[1] "Metro" level corresponds to a specific metropolitan code in urban areas or a Public Use Micro Area (PUMAs) in rural areas, so the geographic coverage is national.

Table 3: Multilevel logit models of the probability of speaking Spanish at home, second generation immigrants, 2014 SIPP data

Row	Variable name	Model (1)	(2)	(3)
1	metro level proportion native born Latino (Metro %NB Latino)	1.740*	0.400	0.468
		(0.799)	(0.955)	(0.971)
2	metro level proportion foreign born Latino (Metro %FB Latino	0.0598	-1.114	-4.619*
		(1.827)	(1.910)	(2.317)
3	tract level proportion native born Latino (Tract %NB)		1.383**	1.170*
			(0.500)	(0.504)
4	tract level proportion foreign born Latino (Tract %FB)		0.945	0.361
			(0.495)	(0.540)
5	tract level proportion native born Latinos who speak Spanish at home (NB Tract Spanish)			1.068**
				(0.398)
6	metro level proportion native born Latinos who speak Spanish at home (NB Metro Spanish)			2.536*
				(1.099)
7	household maximum education: high school	-0.180	-0.147	-0.148
		(0.127)	(0.127)	(0.128)
8	household maximum education: college+	-0.394*	-0.267	-0.241
		(0.153)	(0.156)	(0.156)
	Household composition variables:			
9	number of first generation male children	1.015*	1.034*	1.071*
		(0.423)	(0.426)	(0.430)
10	number of first generation female children	0.368	0.425	0.433
		(0.400)	(0.403)	(0.407)
11	number of first generation adult males	0.785***	0.759***	0.745***
		(0.0861)	(0.0864)	(0.0866)
12	number of first generation adult females	0.959***	0.915***	0.892***
		(0.0894)	(0.0902)	(0.0905)
13	number of 2nd generation male children	-0.0551	-0.0596	-0.0585
		(0.0614)	(0.0619)	(0.0619)
14	number of 2nd generation female children	0.0379	0.0305	0.0292
		(0.0589)	(0.0590)	(0.0591)
15	number of 2nd generation adult males	0.0534	0.0535	0.0660

Table 3: Multilevel logit models of the probability of speaking Spanish at home, second generation immigrants, 2014 SIPP data

Row	Variable name	Model (1)	(2)	(3)
		(0.0908)	(0.0914)	(0.0916)
16	number of 2nd generation adult females	0.181*	0.159	0.160
		(0.0904)	(0.0909)	(0.0910)
17	number of third generation male children	0.109	0.0969	0.101
		(0.0931)	(0.0941)	(0.0947)
18	number of third generation female children	0.241*	0.205*	0.205*
		(0.0966)	(0.0973)	(0.0979)
19	number of third generation adult males	-0.397***	-0.356**	-0.348**
		(0.115)	(0.115)	(0.115)
20	number of third generation adult females	-0.351***	-0.340**	-0.335**
		(0.105)	(0.106)	(0.107)
21	Constant	-0.314	-0.418*	-1.718***
		(0.188)	(0.189)	(0.432)
	Random effect standard deviation	-0.147	-0.149	-0.133
		(0.145)	(0.145)	(0.145)
	Number of cases (rounded to the nearest 10)	3600	3600	3600
	Log likelihood	-1847.0	-1832.0	-1823.0

Notes: Standard errors in parentheses. * -<.05, ** p<0.01, *** p<0.001

Table 4: Multilevel logit models of the probability of speaking Spanish at home, third generation immigrants, 2014 SIPP data

Row	Variable name	Model (4)	(5)	(6)
1	metro level proportion native born Latino (Metro %NB Latino)	1.616 (0.895)	-0.471 (0.991)	-1.516 (0.878)
2	metro level proportion foreign born Latino (Metro %FB Latino	0.168 (2.348)	-1.937 (2.414)	-7.378** (2.381)
3	tract level proportion native born Latino (Tract %NB)		2.162*** (0.426)	1.647*** (0.433)
4	tract level proportion foreign born Latino (Tract %FB)		2.033** (0.623)	1.123 (0.666)
5	tract level proportion native born Latinos who speak Spanish at home (NB Tract Spanish)			1.470*** (0.410)
6	metro level proportion native born Latinos who speak Spanish at home (NB Metro Spanish)			5.214*** (1.163)
7	household maximum education: high school	-0.323* (0.157)	-0.252 (0.157)	-0.204 (0.156)
8	household maximum education: college+	-0.525** (0.174)	-0.253 (0.177)	-0.209 (0.176)
	Household composition variables:			
9	number of first generation adult males	0.296 (0.189)	0.200 (0.190)	0.175 (0.188)
10	number of first generation adult females	0.561** (0.197)	0.541** (0.201)	0.530** (0.198)
11	number of 2nd generation male children	0.224 (0.159)	0.251 (0.161)	0.250 (0.159)
12	number of 2nd generation female children	0.275 (0.170)	0.226 (0.176)	0.230 (0.175)
13	number of 2nd generation adult males	0.770*** (0.133)	0.719*** (0.134)	0.730*** (0.133)
14	number of 2nd generation adult females	0.425*** (0.122)	0.380** (0.125)	0.381** (0.123)

Table 4: Multilevel logit models of the probability of speaking Spanish at home, third generation immigrants, 2014 SIPP data

Row	Variable name	Model (4)	(5)	(6)
15	number of third generation male children	0.0237 (0.0588)	0.0196 (0.0599)	0.0180 (0.0594)
16	number of third generation female children	-0.0864 (0.0605)	-0.113 (0.0613)	-0.113 (0.0609)
17	number of third generation adult males	0.181** (0.0616)	0.174** (0.0626)	0.173** (0.0620)
18	number of third generation adult females	0.0929 (0.0674)	0.131 (0.0685)	0.124 (0.0680)
19	Constant	-1.727*** (0.247)	-1.974*** (0.247)	-4.165*** (0.465)
20	Random effect standard deviation	0.373*** (0.103)	0.356*** (0.104)	0.132 (0.121)
21	Number of cases (rounded to the nearest 100)	3300	3300	3300
22	Log likelihood	-1819.0	-1778.0	-1755.0

Notes: Standard errors in parentheses. * -<.05, ** p<0.01, *** p<0.001